

GTIIT Spring 2025 Calculus 1M

Workshop 7

April 21, 2025

Try to solve the following exercises. Feel free to chat about them with your classmates, but make sure to write your own solutions. If you get stuck, just move on to the next one and come back later.

If you need any help at all —whether it's a tiny hint or a big clue— don't hesitate to ask me!

Important note: If you find any errors or typos, please let me know so I can correct them.

1. Recall all the guide lines to sketch the graph of a function.
2. Sketch the graph of the following functions (hard):
 - (i) $y = \frac{1}{x^2-4}$
 - (ii) $y = \frac{1}{x} + \ln x$
 - (iii) $y = \ln(x^2 + c)$, where c is a constant.
3. Show that of all the isosceles triangles with a given perimeter, the one with the greatest area is equilateral.
4. Find the dimensions of the rectangle of largest area that can be inscribed in an equilateral triangle of side L if one side of the rectangle lies on the base of the triangle.
5. Find the antiderivative of the following functions:
 - (i) $f(x) = 2x^3 - 3x^2 + 4$
 - (ii) $f(x) = \frac{1}{x^2}$
 - (iii) $f(x) = \sin(2x)$
 - (iv) $f(x) = e^{3x}$